

G. Aravamudan
J. Gopalakrishnan
and M. R. Udupa

Indian Institute of Technology
Madras 36, India

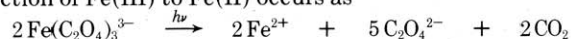
Preparation and Properties of Potassium Trioxalatoferrate(III) Trihydrate

A laboratory exercise

Recently,¹ a simple method for the preparation of potassium trioxalatoferrate(III) trihydrate has been reported as an illustrative exercise in general chemistry laboratory courses. We have been giving an exercise involving the title compound to graduate students of chemistry not only from the preparative aspects but also from the point of view of structure and reactivity of coordination compounds. The exercise involves preparation, characterization, and study of spectral, magnetic, photochemical, and thermal behavior of the title compound.

The compound is stable and can be prepared in a state of high purity from easily available chemicals.¹ It is a good example of a high spin iron(III) complex with D_{3h} configuration of ligands around the metal ion. The ir absorption spectrum of the solid shows that oxalato groups (which can ligate variously) are present as chelating bidentate entities. The electronic spectrum of the solid and of its solution in dilute acid media are also recorded and interpreted. Whereas $d-d$ transitions in the high spin d^5 system are weak, the moderately intense green color of the solid and the yellow color of its solution in acid medium are characteristic of charge transfer transition from oxalate to iron(III). The magnetic moment μ_{eff} at room temperature determined by the Gouy method is almost equal to the "spin only" value of the d^5 system in keeping with the high spin configuration of iron(III) in the compound.² The large value of the magnetic moment is an added advantage in an introductory exercise.

The complex affords an interesting comparison between the thermal and photochemical stabilities^{3,4} of oxalato complexes of iron(III). In the dark, solutions of the complex in acid medium ($\sim 2\text{ N H}_2\text{SO}_4$) are stable even up to 100°C . However, when the same solution is exposed to bright sunlight for 5–10 min or to radiation from a 400-W mercury-vapor lamp for 15–20 min, quantitative photoreduction of Fe(III) to Fe(II) occurs as



The completion of the photoreduction is indicated by the disappearance of the yellow color. Fe(II) thus formed is determined by titration against sodium vanadate(V) using diphenylamine indicator in the presence of phosphoric acid, thus affording a simple method for quantitative determination of iron in the original complex. The students are made aware that conventional oxidants such as KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$ cannot be employed for the determination of iron(II), since the remaining oxalate in solution would react with such strong multi-equivalent oxidants in the presence of Fe(II) even in the cold. The reasons for induced reactions occurring in MnO_4^- (or $\text{Cr}_2\text{O}_7^{2-}$) - Fe^{2+} - $\text{C}_2\text{O}_4^{2-}$ systems and their absence in the vanadate- Fe^{2+} - $\text{C}_2\text{O}_4^{2-}$ system are explained to the students.

The thermogravimetric curve, obtained with 500 mg of the sample in air reveals that the compound loses three molecules of water around 110°C and the anhydrous compound undergoes thermal reduction around 260°C to an oxalate of iron(II), identified to be $\text{K}_6[\text{Fe}_2(\text{C}_2\text{O}_4)_5]$ which subsequently undergoes oxidative decomposition around 400°C to a mixture of K_2CO_3 and Fe_2O_3 .⁵

The preparation and analysis of the compound comprise one laboratory experiment of 6-hr duration, while the study of magnetic spectral, photochemical, and thermal behavior forms another exercise of equal duration. All the students are able to complete the exercise within the specified time with excellent results.

¹ Johnson, R. C., J. CHEM. EDUC., 47, 702 (1970).

² Earnshaw, A., "Introduction to Magnetochemistry," Academic Press, London, 1968, pp. 34 and 57.

³ Balzani, V., and Carassiti, V., "Photochemistry of Coordination Compounds," Academic Press, London, 1970, p. 167.

⁴ Rao, G. G., Aravamudan, G., and Venkatamma, V. C., Z. Anal. Chem., 146, 161 (1955).

⁵ Bancroft, G. M., Dharmawardena, K. G., and Maddock, A. G., Inorg. Chem., 9, 223 (1970).